KP13 Generalized Linear Classifiers

> Level: undergraduate, year 3
>
> Pattern Recognition | Chapter 4, §4.12

Slides: read and download

Learning objectives

- Explain why we lift first and then use a linear classifier, and how a nonlinear boundary in the original space becomes a plane in the new space.
- Compute the weights of a generalized linear classifier by least squares (the normal equation) and handle several classes.
- Explain how a generalized linear classifier differs from projection pursuit and from a multilayer perceptron, and the price of lifting.

KP13 Generalized Linear Classifiers

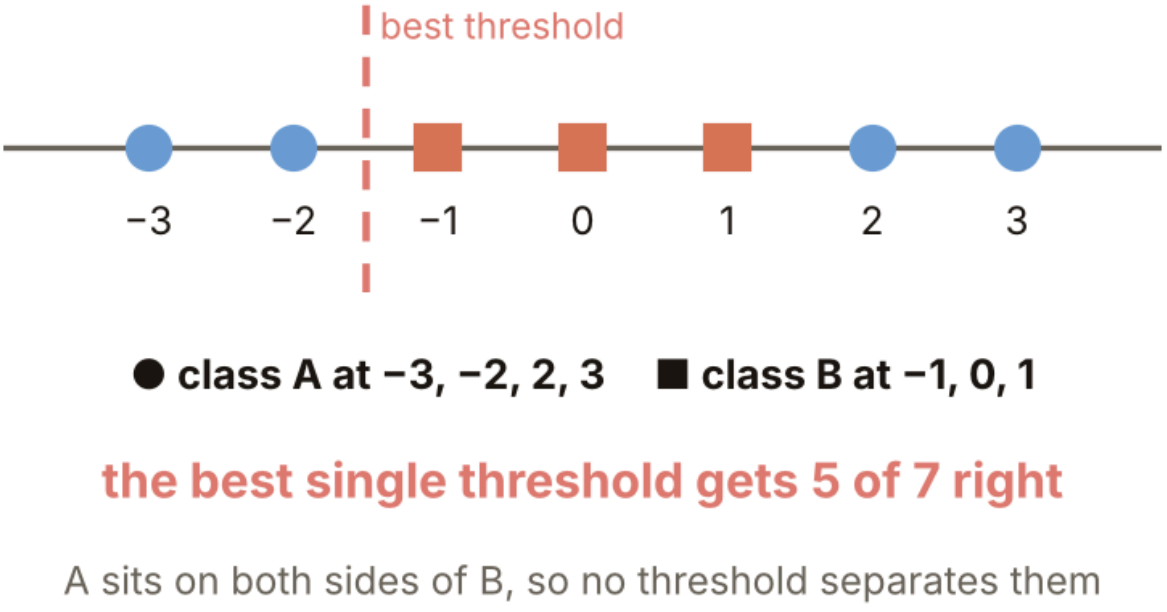
One-sentence summary: First send x to a k-dimensional space with a set of fixed nonlinear functions f_i, then design an ordinary linear classifier there; all the nonlinearity is carried by the f_i and the discriminant function stays linear in the weights.

1. Sandwiched classes

Plain explanation: What if no straight line separates the data in the original space?

Example (one dimension, designed here): class A at $x = -3, -2, 2, 3$ and class B at $x = -1, 0, 1$. Any single threshold, in either direction, gets at most 5 of the 7 right (exhaustive program search). The reason is that A lies on both sides of B, so x alone cannot show it.

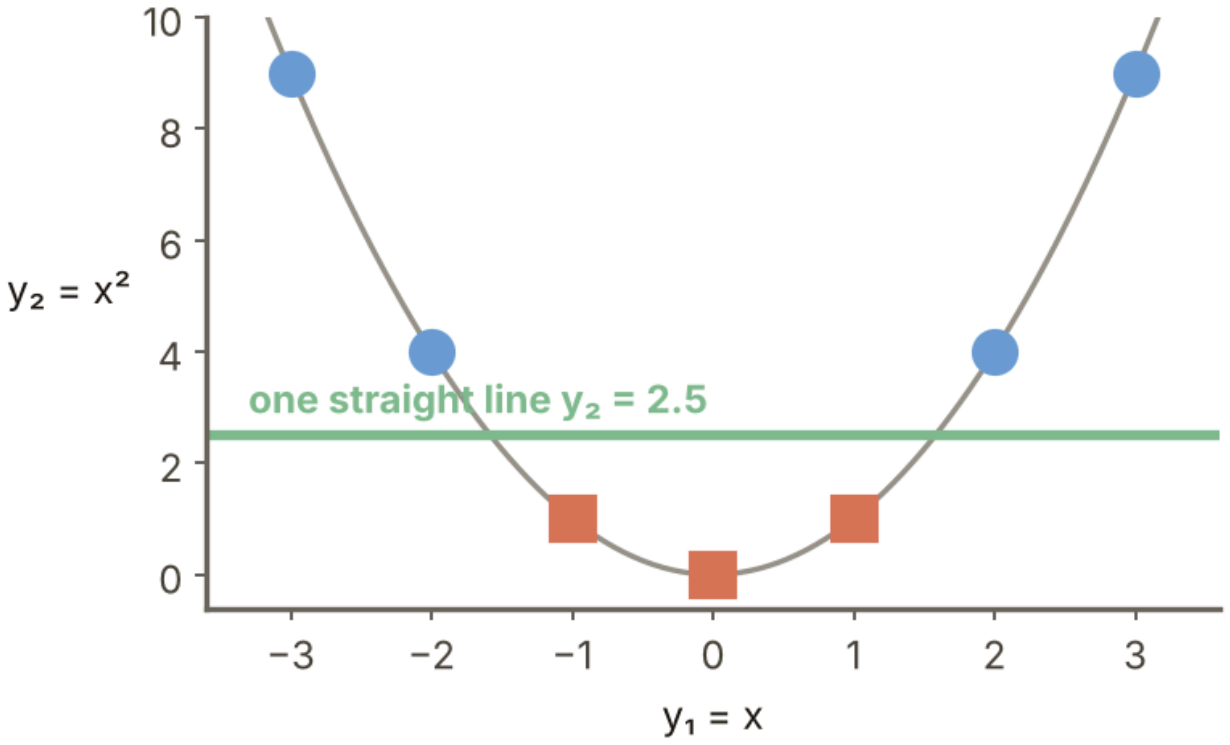
Figure `lec05-kp13-sandwich`: the two classes on a line and the best single threshold.



2. A new coordinate

Feature mapping: use a set of functions f_i to replace x by new coordinates $y = (f_1(x), \ldots, f_k(x))$. Here $y = (x, x^2)$: A has $y_2 = 4, 9$ and B has $y_2 = 0, 1$, and one horizontal line $y_2 = 2.5$ separates the two classes, a linear classifier in the new coordinates.

Figure `lec05-kp13-lift`: the points lifted to (x, x^2) and separated by one horizontal line.



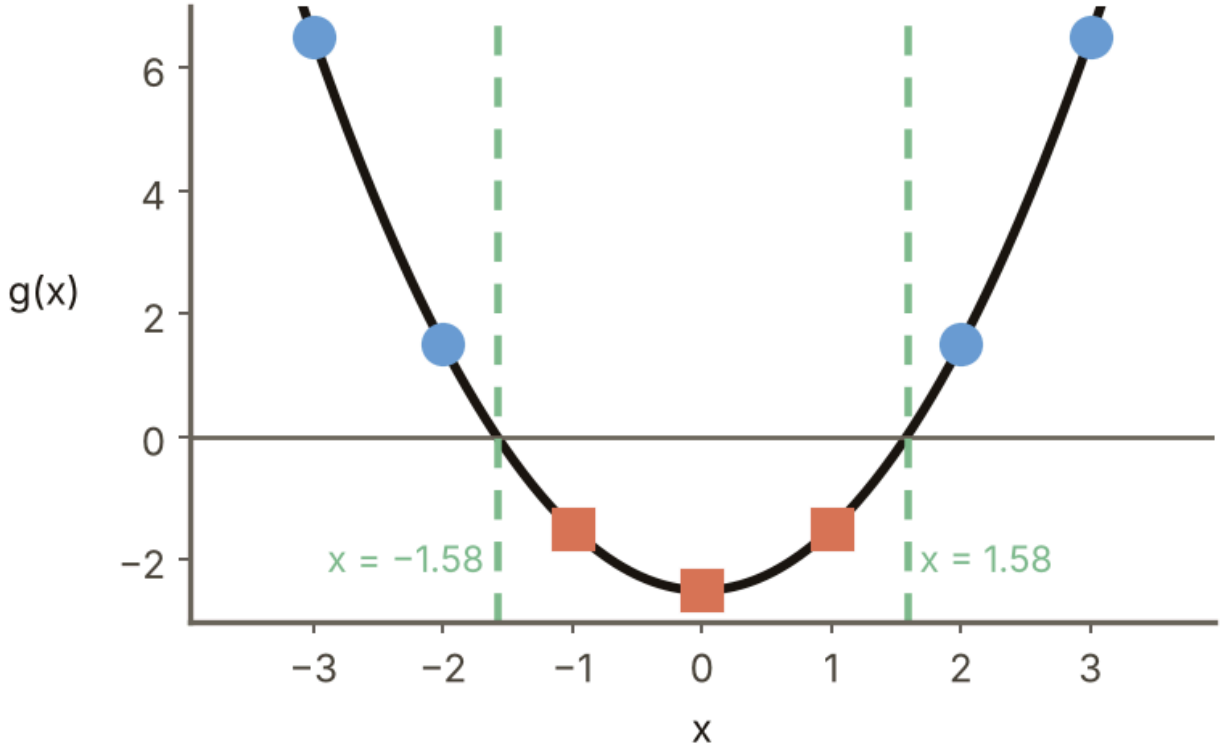
3. Curved when seen back on x

Formal definition: the **generalized linear classifier**

$$g(x) = w_0 + \sum_{i=1}^{k} w_i f_i(x),$$

lifts with fixed nonlinear f_i and then designs an ordinary linear classifier; "linear" means linear in the weights w_i. Here $g(x) = -2.5 + 0 \cdot x + 1 \cdot x^2$, $g > 0$ means A and $g < 0$ means B; $g = 0$ at $x = \pm\sqrt{2.5} = \pm 1.58$, A outside and B inside. A line (in general a hyperplane) in the new space is two points in the original space. Seen the other way round, if a nonlinear hypersurface $g(x) = 0$ separates the classes in the original space, writing g as a linear combination of the f_i is the classical interpolation and function-approximation problem (source).

Figure `lec05-kp13-curve`: $g(x) = -2.5 + x^2$ and the seven points.



4. XOR once more

Add the coordinate $y_3 = x_1 x_2$ and take $g = x_1 + x_2 - 2y_3 - 0.5$: $(0,0) \to -0.5$, $(0,1) \to +0.5$, $(1,0) \to +0.5$, $(1,1) \to -0.5$, so one plane separates the four points. KP02 combined two lines; here one fixed mapping and one plane do the job.

Figure `lec05-kp13-xor`: new coordinates and g of the four XOR points.

x_1	x_2	$y_3 = x_1 x_2$	g	class
0	0	0	−0.5	B
0	1	0	0.5	A
1	0	0	0.5	A
1	1	1	−0.5	B

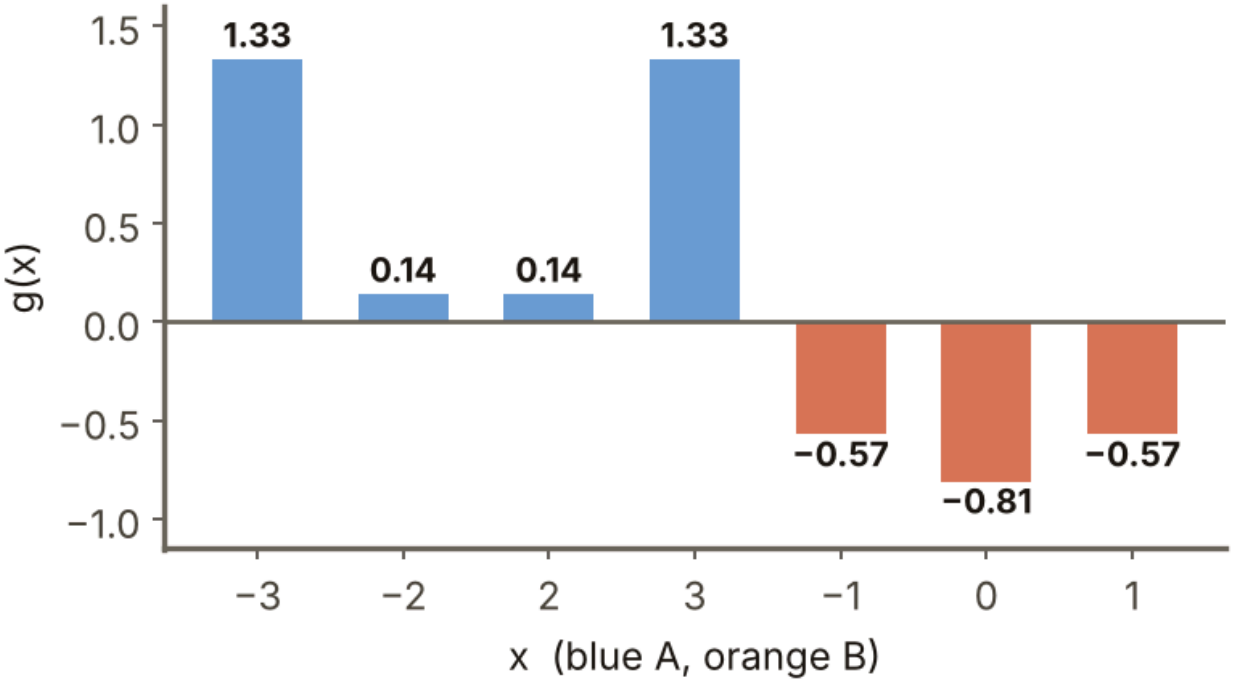
$$g = x_1 + x_2 - 2 \cdot y_3 - 0.5$$

a plane in (x_1, x_2, y_3): the XOR points are now separable

5. Computing the weights

Normal equation: the solution $w = (\Phi^T \Phi)^{-1} \Phi^T t$ that minimizes the squared error between Φw and the targets ($+1$ for A, -1 for B); each row of Φ is $(1, f_1, \ldots, f_k)$ of one sample. For the seven points of section 1 the rows of Φ are $(1, x, x^2)$ and the solution is $w = (-0.810,\ 0,\ 0.238)$; the sign of each of the seven g equals its target, so all seven points are right (program).

Figure `lec05-kp13-fit`: the g values of the seven points.



least-squares weights, targets +1 (A) and −1 (B)

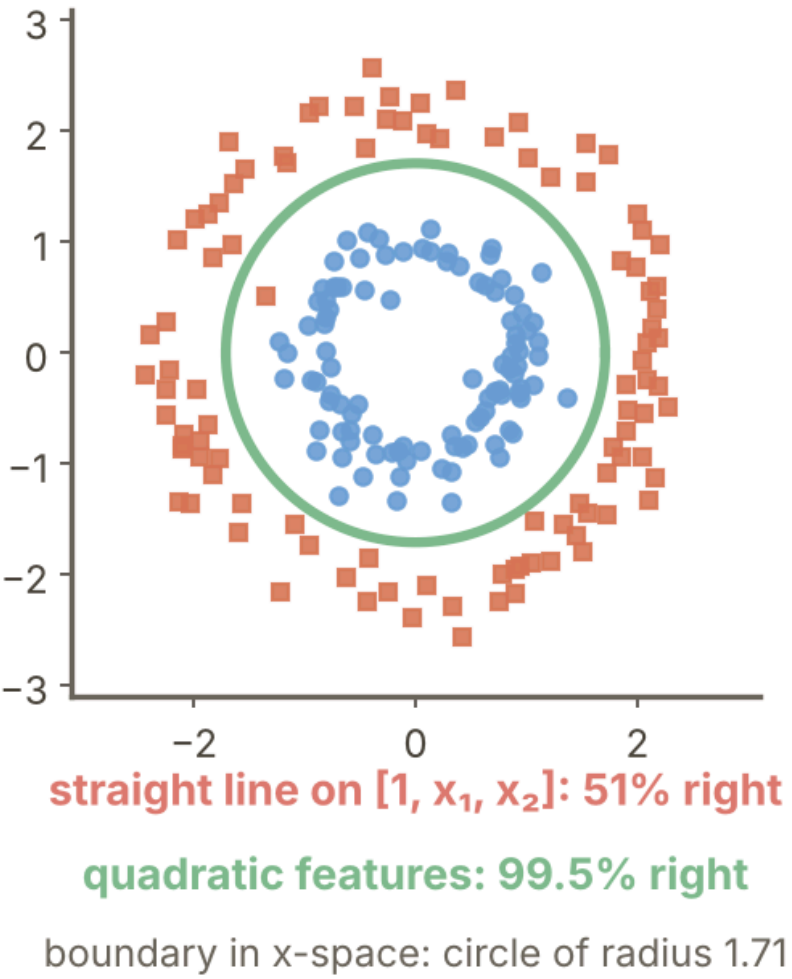
w = (−0.810, 0.000, 0.238)

every g has the sign of its target: 7 of 7 right

6. Concentric rings

A program generates two classes of 100 points each, the inner ring of radius about 1.0 and the outer about 2.2. Straight-line features $[1, x_1, x_2]$ are right only 51% of the time, no better than guessing; quadratic features $[1, x_1, x_2, x_1^2, x_2^2, x_1 x_2]$ are 99.5% right, and $g = 0$ is a circle of radius 1.71 in the original space, between the two rings (program). **Common misconception**: after lifting the boundary becomes a straight line. It is a plane in the new space and a circle back in the original space.

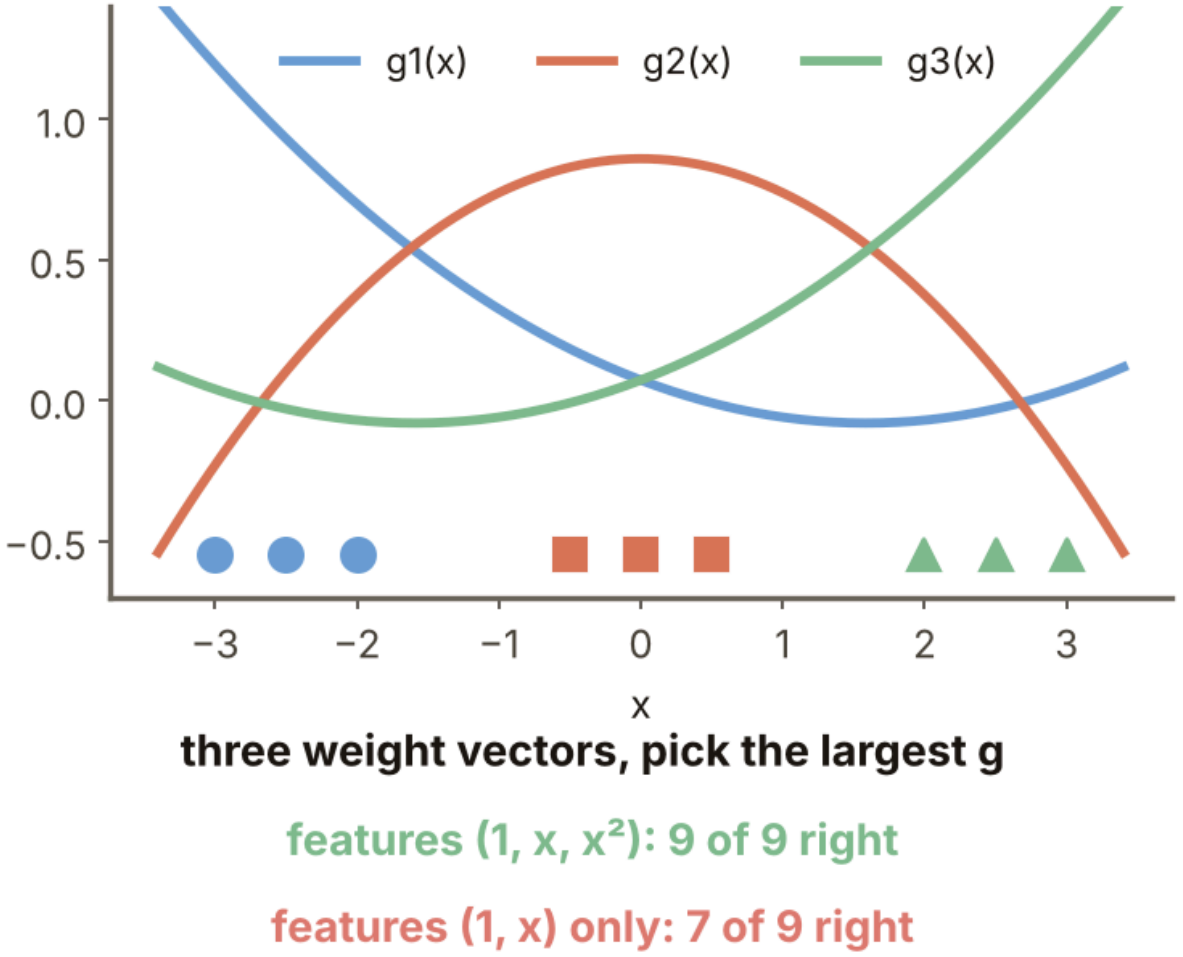
Figure `lec05-kp13-rings`: the concentric rings and the decision circle.



7. Several classes

For M classes design M weight vectors w_r and pick the class with the largest $w_r^T y + w_{r0}$. **Example** (designed here): nine points in one dimension with classes 1, 2, 3 on the left, middle and right; one-hot targets, features $(1, x, x^2)$ and a least-squares solution for three weight vectors: all nine right; with only $(1, x)$ just 7 are right and the middle class is right only at $x = 0$ (a three-way tie of $1/3$ there, resolved to the middle class).

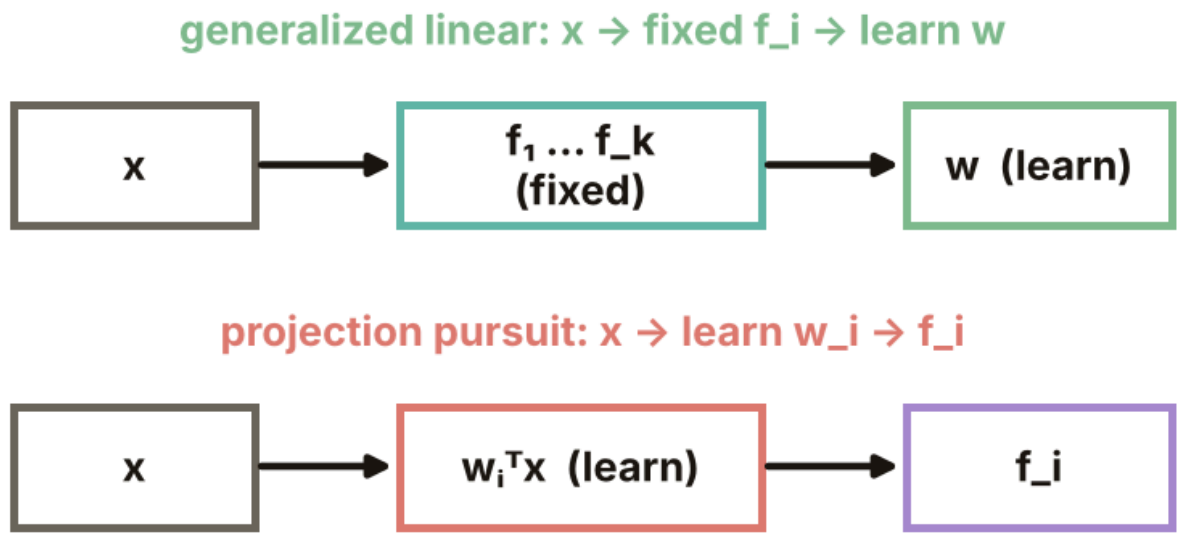
Figure `lec05-kp13-multi`: the three g curves and nine points.



8. Contrast: projection pursuit

Projection pursuit: $g(x) = \sum_i f_i(w_i^T x)$; f_i sees the projection of x on a direction w_i, and direction and function are optimized together. Double w: a generalized linear g doubles exactly, while for projection pursuit (f a sigmoid, input 1, $w = 1$) $f(2) = 0.881$, not $2f(1) = 1.462$, so it is not linear in w and needs a two-stage alternating optimization (source). If f_i is fixed to a sigmoid it is a one-hidden-layer neural network, which multilayer perceptrons have replaced in practice. **To remember**: generalized linear = fixed mapping + learnable linear layer; multilayer perceptron = learnable mapping + learnable linear layer.

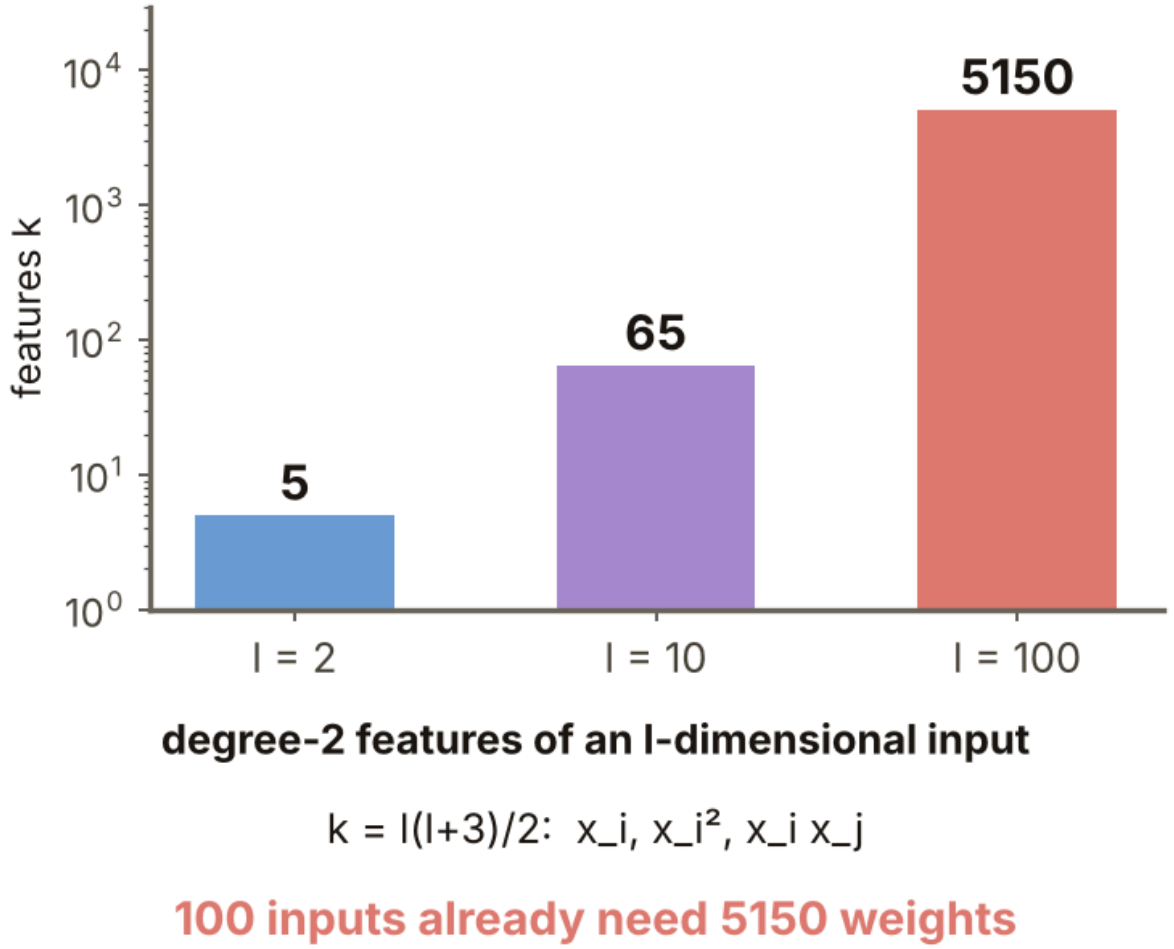
Figure `lec05-kp13-pp`: block diagrams of the two structures and the doubling test.



Extension: how many coordinates, practice and worked solution

How many coordinates: quadratic features of an l-dimensional input are l terms x_i, l terms x_i^2 and $l(l-1)/2$ terms $x_i x_j$, $k = l(l+3)/2$ in all: 5 for $l = 2$, 65 for $l = 10$ and 5150 for $l = 100$. Does lifting really help? The source answers with Cover's theorem (KP14); polynomial and RBF mappings follow in KP15 and KP16.

Figure `lec05-kp13-count`: the number of quadratic features (log axis).



Practice: features $y = (x_1^2, x_2^2)$ and $g = y_1 + y_2 - 4$ ($g > 0$ is A).

1. Which class are $(1, 1)$, $(3, 0)$ and $(0, -2.5)$?
2. What shape is $g = 0$ on the x plane?

Figure `lec05-kp13-practice`: the setup of the practice.

features $y = (x_1^2, x_2^2)$

$g = y_1 + y_2 - 4$

$g > 0 \rightarrow$ class A, $g < 0 \rightarrow$ class B

classify (1, 1), (3, 0), (0, −2.5)

and describe the boundary g = 0 in the x-plane

Worked solution of questions 1 and 2: (1) $(1, 1)$: $1 + 1 - 4 = -2$, B; $(3, 0)$: $9 + 0 - 4 = 5$, A; $(0, -2.5)$: $0 + 6.25 - 4 = 2.25$, A. (2) $g = 0$ means $x_1^2 + x_2^2 = 4$, a circle of radius 2: a straight line in the y space and curved in the x space.

Figure `lec05-kp13-practice-sol`: g of the three points and the decision circle.

$(1, 1)$: $1 + 1 - 4$ $= -2 \rightarrow B$

$(3, 0)$: $9 + 0 - 4$ $= +5 \rightarrow A$

$(0, -2.5)$: $0 + 6.25 - 4$ $= +2.25 \rightarrow A$

$$g = 0 \Leftrightarrow x_1^2 + x_2^2 = 4$$

a circle of radius 2: a plane in y, curved in x

Terms and sources

English	中文
Feature mapping	特徵映射
Generalized linear classifier	廣義線性分類器
Normal equation	正規方程
Projection pursuit	投影追蹤

Source: Chapter 4 KP13, §4.12 (read 2026-10-25). The definition of the generalized linear classifier, the function-approximation view, the several-class method, the projection-pursuit contrast and the "to remember" line follow the source; the one-dimensional sandwich example, the XOR coordinates, the concentric-rings data (the source has an interactive demo, this lecture generates its own data by a program), the three-class one-dimensional data and every number were designed and recomputed for this lecture ([added], (program)).